

BRIEF REPORT

Sex-Specific Cardiovascular Strain During Heat Acclimation in Tactical Athletes



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Tactical athletes (TAs) including military personnel, firefighters, and law enforcement officers bear a disproportionate burden of cardiovascular disease compared with age-matched civilian populations.¹ Repeated occupational exposure to extreme physical exertion and environmental stressors may contribute to this elevated risk by increasing cardiovascular strain and predisposing individuals to acute cardiac events during duty-related activities.² As women represent a growing proportion of the TA workforce, there is an increasing imperative to understand sex-specific optimization of cardiovascular health and readiness.

Exercise in hot environments markedly augments cardiovascular demand, requiring large increases in cardiac output to simultaneously

support thermoregulation and skeletal muscle perfusion. These demands are met through elevations in heart rate (HR) and myocardial contractility, resulting in increased myocardial workload and oxygen demand. Collectively termed cardiovascular strain in the exercise physiology literature, this hemodynamic burden on the heart is distinct from myocardial deformation (strain) as assessed by echocardiography. This increased myocardial work during prolonged exertion in the heat may be a physiological contributor to acute cardiovascular events, particularly in occupational athletes.¹

Exercise heat acclimation (HA) attenuates the physiological stress associated with exercise-heat exposure.^{2,3} However, the impact of HA on indices of cardiovascular strain relevant to cardiovascular risk, and whether these adaptations differ between women and men, are poorly understood. We evaluated changes in cardiovascular strain during exercise in the heat across a 10-day HA protocol in male and female TAs.

METHODS

This study is a secondary analysis of a controlled work-rate HA investigation in healthy adults, approved by the U.S. Army Medical Research and Development Command Institutional Review Board (M-10929) and registered at ClinicalTrials.gov (NCT05292170). All participants provided written informed consent per the Declaration of Helsinki.

What is the clinical question being addressed?

Does heat acclimation reduce cardiovascular strain during exercise in the heat, a physiological contributor to acute cardiovascular risk, in male and female tactical athletes?

What is the main finding?

Ten days of heat acclimation reduced heart rate and rate-pressure product in both sexes, indicating lower cardiovascular strain and supporting heat acclimation as a strategy to mitigate cardiovascular risk in tactical athletes.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS**

BP = blood pressure
HA = heat acclimation
HR = heart rate
RPP = rate-pressure product
TA = tactical athlete

Full methodological details are published in the parent study.⁴

Briefly, 27 healthy volunteers (16 females) completed 10 HA visits over 14 days (first 4 consecutive; ≤ 1 consecutive rest day thereafter), exercising on a treadmill in an environmental chamber (40 °C, 40% relative humidity, 1.34 m s⁻¹ wind speed) at a fixed work rate (5 km h⁻¹, 2% grade) for 120 minutes with ad libitum hydration. Testing occurred October–April to minimize seasonal acclimatization.

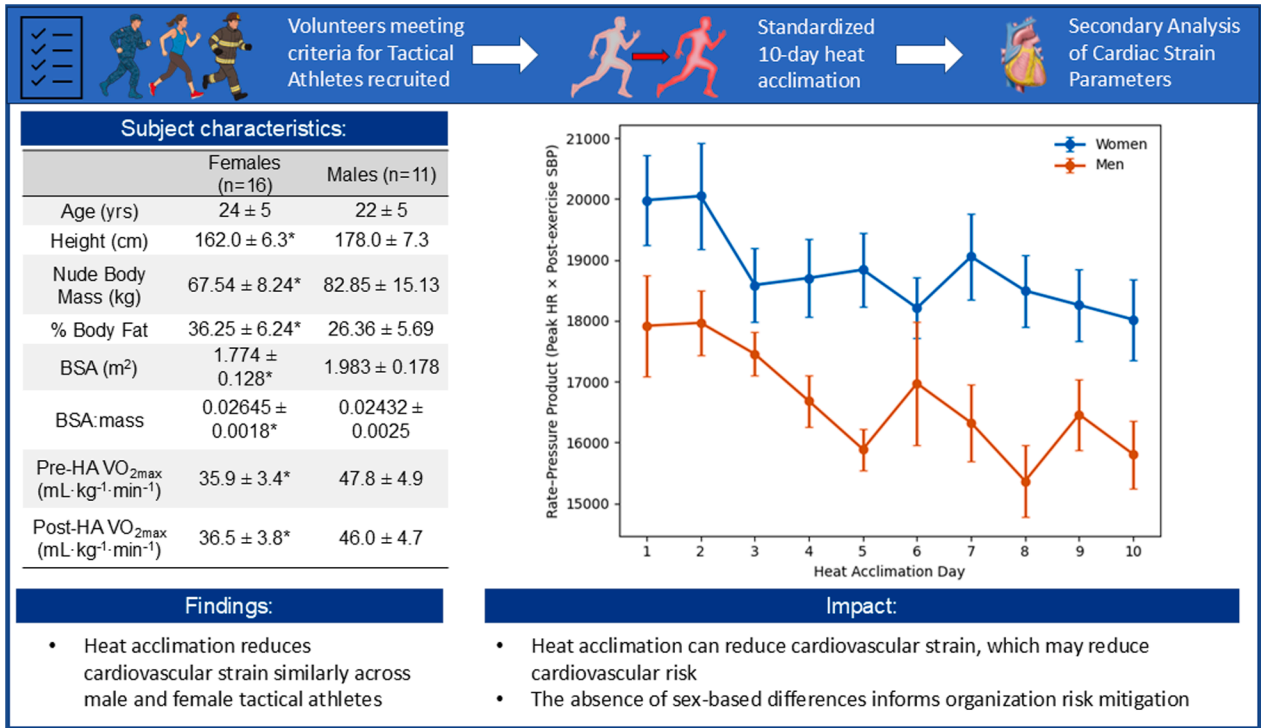
The primary outcomes for this analysis were exercise HR, postexercise blood pressure (BP), and rate-pressure product (RPP), assessed across the 10-day HA protocol. RPP was calculated as the product of peak exercise HR and immediate postexercise systolic BP, and is a well-validated, noninvasive index of myocardial oxygen demand used in clinical exercise testing. HR was measured continuously via telemetric monitor throughout each 120-minute exercise session. BP was assessed by auscultation in the seated position immediately upon cessation of exercise.

Statistics: Linear mixed-effects models were used to assess changes over the 10-day HA protocol and to evaluate sex differences, consistent with the parent study.⁴ Models included fixed effects for sex, time (categorical), their interaction, and body surface area to mass ratio; subject-level and time-within-subject random effects were also included. Parameters were estimated via restricted maximum likelihood (Kenward-Roger method). Marginal effects of sex were calculated at each time point; day 1 vs day 10 comparisons were assessed within each sex. Analyses used Stata SE 18.0; significance was set at $P < 0.05$.

RESULTS

Twenty-seven individuals (16 females, 11 males) completed the study (Figure 1). The mean ages were 24 ± 5 years (females) and 22 ± 5 years (males). Females were shorter, lighter, had higher body fat percentage, larger body surface area to mass ratio, and lower maximal oxygen consumption than males (all $P < 0.05$) (Figure 1).

FIGURE 1 Sex-Specific Cardiovascular Strain During Heat Acclimation in Tactical Athletes



Rate-pressure product (RPP), calculated as the product of peak exercise heart rate and immediate postexercise systolic blood pressure, is shown for each day of the 10-day heat acclimation (HA) protocol for women (n = 16) and men (n = 11). RPP, as a noninvasive index of myocardial oxygen demand and a standard measure of cardiovascular strain in exercise physiology, decreased progressively from day 1 to day 10 of HA. The magnitude of reduction was similar between women and men. Data are represented as mean ± SD. Linear mixed-effects models were used to calculate differences between and within groups. Subject characteristics are provided on the left. *Denotes significant differences between sexes ($P < 0.05$). BSA = body surface area; BSA:mass = body surface area to mass ratio; HR = heart rate; SBP = systolic blood pressure; VO_{2max} = maximal oxygen consumption.

HR during exercise was higher in women compared with men throughout ($P < 0.001$ at each time point) and decreased similarly over time in both sexes. Peak exercise HR declined significantly from day 1 to day 10 in men (141 ± 16 – 127 ± 11 beats/min, $P < 0.001$) and women (170 ± 17 – 152 ± 17 beats/min, $P < 0.001$), with no significant sex-by-time interaction ($P > 0.05$).

Postexercise arterial BP remained stable across HA in both groups. Mean postexercise systolic, diastolic, and mean arterial pressures were 126.9 ± 2.6 , 68.5 ± 3.2 , and 87.9 ± 2.2 mm Hg in men, and 121.3 ± 2.8 , 71.9 ± 2.6 , and 88.4 ± 2.2 mm Hg in women, respectively.

RPP decreased significantly from day 1 to day 10 in both women ($-1,949 \pm 2,027$, $P = 0.003$) and men ($-2,111 \pm 1,954$, $P = 0.005$), with no significant difference in the magnitude of reduction between sexes (Figure 1). As workload remained constant, there was no significant relationship between workload achieved and reduction in RPP in either sex.

DISCUSSION

The primary finding of the present analysis is that 10 days of controlled-work HA significantly reduced cardiovascular strain during exercise in both men and women. This was evidenced by marked reductions in peak exercise HR and RPP, indicating a lower myocardial workload at a fixed external workload following HA (Figure 1). These adaptations both decrease risk of adverse cardiovascular events and increase work capacity. Importantly, the similarity in adaptation between men and women supports the idea that the overall clinical risk associated with exercise-heat exposure is not greater in women, when similar work is performed in groups of healthy young people.⁴

RPP is a well-established index of myocardial oxygen demand and has long been used in clinical cardiology to assess cardiovascular stress during exercise. The observed reduction in exercise RPP with HA therefore represents a physiologically meaningful decrease in cardiac work during conditions that closely approximate operational heat exposure. Although women demonstrated higher HR throughout HA, the magnitude of cardiovascular adaptation was similar between sexes.

The mechanisms underlying reduced cardiovascular strain with HA are likely multifactorial and include plasma volume expansion, enhanced peripheral vasodilation, and increased sweating. HA has also been shown to increase expression of heat shock proteins, which may further enhance

myocardial and vascular resilience during thermal stress.⁵ Interestingly, in the present subject population, the increase in heat shock protein 72 was similar between men and women, further supporting the idea of similar adaptability and resilience between groups.⁴

Our findings have important implications from both clinical and operational perspectives. Acute cardiovascular events in TAs frequently occur during strenuous activity in hot environments,³ and interventions that lower cardiovascular strain during exertion may reduce susceptibility to such events. HA represents a practical, nonpharmacologic strategy that improves physiological tolerance to heat stress while potentially mitigating cardiovascular risk. The absence of sex-based differences in cardiovascular adaptation suggests that uniform HA protocols can be applied across tactical populations, aligning clinical risk mitigation with organizational preparedness goals.

The limitations of the present study are 2-fold. First, as described previously, this was a secondary analysis of previously published work that did not have cardiovascular function as its main outcome;⁴ thus, direct clinical cardiovascular measures were not included. However, the use of rate-pressure product provides a clinically relevant surrogate of myocardial workload during exercise. Second, HR and arterial pressure values were not measured at the same time. However, due to the low-intensity nature of the exercise, it was likely that BP was similar during exercise compared to seated values immediately afterward.

We conclude that HA substantially, and similarly, reduces cardiovascular strain during exertion in the heat in both women and men, supporting its role as a key component of cardiovascular risk mitigation and readiness strategies in TAs.

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REFERENCES

1. Murphy CE, Dean JHL, Dineen EH, Haigney MC. Exercise at the extremes: toward an understanding of increased cardiovascular risk in the tactical athlete. *Curr Treat Options Cardiovasc Med*. 2025;27(1):25.
2. Giersch GEW, Bradbury KE, Charkoudian N. Optimizing female warfighter health and performance in environmental extremes. *Med Surveill Mon Rep*. 2022;32(5):35–38.
3. Périard JD, DeGroot D, Jay O. Exertional heat stroke in sport and the military: epidemiology and mitigation. *Exp Physiol*. 2022;107(10):1111–1121.
4. Giersch GEW, Greenfield AM, Brazelton SC, et al. Distinct physiological responses to heat acclimation in males and females lead to similar thermal adaptations in both sexes. *J Appl Physiol*. 2025;139(3):863–872.
5. McClung JP, Hasday JD, He JR, et al. Exercise-heat acclimation in humans alters baseline levels and ex vivo heat inducibility of HSP72 and HSP90 in peripheral blood mononuclear cells. *Am J Physiol Regul Integr Comp Physiol*. 2008;294(1):R185–R191.

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